

## Curriculum Vitae

### Profile

Undergraduate student at Xingjian College, Tsinghua University, pursuing a dual-degree track in Theoretical and Applied Mechanics and Energy and Power Engineering. Expected graduation: July 2028. Focus areas include AI application engineering, RAG / GraphRAG, engineering simulation, product prototyping, and 3D parametric modeling.

### Education

2024.09–2028.07 **B.Eng. candidate, Tsinghua University, Xingjian College, Beijing, Expected**  
Dual-degree track: Theoretical and Applied Mechanics + Energy and Power Engineering. Completed 24 courses, including 17 required courses.

### Skills

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|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| AI engineering | RAG / GraphRAG exploration, knowledge-base construction, vector search, backend control panels, and engineering validation. |
| Simulation     | COMSOL modeling, Python numerical simulation, engineering-result reproduction, and parametric experiments.                  |
| Product        | Requirement analysis, interaction flows, agent prototyping, AIGC 3D generation, and presentation narrative.                 |
| Workflow       | Tool-first research, context engineering, multi-agent collaboration, human review, and staged verification.                 |

### Selected Projects

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| RAG / GraphRAG         | <b>Power-equipment knowledge-base console, FastAPI, Chroma, vector search</b> <ul style="list-style-type: none"><li>○ Built a console for upload, ingestion, retrieval, benchmark, and log review, using FastAPI and Chroma to manage chunks, indexes, and runtime data.</li><li>○ Processed 6,098 pages of power-equipment materials into an auditable pipeline and generated 593 chunks for RAG / GraphRAG experiments.</li><li>○ Demo: javonloong.github.io/RAG; Source: github.com/JavonLoong/RAG.</li></ul>             |
| Simulation             | <b>High-altitude wind-energy AWES and COMSOL reproduction, COMSOL, Python, numerical validation</b> <ul style="list-style-type: none"><li>○ Reproduced representative flight trajectories, tether tension, power curves, and pumping-cycle energy accounting.</li><li>○ Connected COMSOL aerodynamic polar data to a Python main simulation and organized a 42-case review loop.</li></ul>                                                                                                                                   |
| AI Agent / AIGC        | <b>Smart electronic-product innovation course project, Agent prototype, interaction flow, 3D generation</b> <ul style="list-style-type: none"><li>○ Completed agent concept design, requirement decomposition, interaction-flow design, and functional prototyping on the iFLYTEK Xingchen platform.</li><li>○ Used Tencent Hunyuan 3D for model generation and refinement; received course awards for Agent design and 3D model design.</li></ul>                                                                           |
| Visual communication   | <b>Aesthetic-education image and presentation production, Image generation, layout control, presentation narrative</b> <ul style="list-style-type: none"><li>○ Produced presentation materials covering asset selection, image style, page redraw, and final delivery.</li><li>○ Focused on visual judgment, information hierarchy, layout stability, and reviewable deliverables.</li></ul>                                                                                                                                 |
| 3D parametric modeling | <b>MoonCake Studio mooncake-mold designer, Three.js, real-time preview, STL / GLB export</b> <ul style="list-style-type: none"><li>○ Implemented a browser-based 3D mooncake-mold customization tool with edge profiles, patterns, Chinese engraving, image relief, and dual-view preview.</li><li>○ Supported STL / GLB export, covering 3D modeling, interaction design, and production-oriented file output.</li><li>○ Demo: javonloong.github.io/Mooncake-Modle; Source: github.com/JavonLoong/Mooncake-Modle.</li></ul> |

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## Leadership and Public Service

- 2025.05–Present **Vice President**, *Tsinghua University Zhi Shi Zhe Association*
- Helped host Bolivian Ambassador to China Hugo Siles and coordinated a demonstration-cooking event for cross-cultural exchange.
  - Proposed and implemented activity plans to improve member engagement.
- 2025 **Participant and logistics lead**, *Xingjian College activities and “Ha-Guo Jian-Xing” overseas practice*
- Participated in student-festival prop work, skating-club teaching, and collective training organization.
  - During the Kazakhstan overseas practice, handled English meeting notes, SDG questionnaire distribution, data collection, insurance reimbursement, transport coordination, and urgent logistics.
- 2024–2025 **Volunteer, public speaker, and tutor**, *Campus service and teaching*
- Served in admissions volunteering, campus guiding, “Return to Alma Mater” visits, Hongyan training-camp service, and large exhibition volunteering.
  - Tutored students in mathematics, physics, Chinese, and full-subject review, emphasizing clear decomposition of complex material.

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## Honors and Awards

- Scholarships Tsinghua University Strong Foundation Plan A+ rating; 2024 Haichang Intelligent Scholarship First Prize; 2024 Dushulang Yuanmeng Scholarship; 2025 National Inspirational Scholarship; 2025 Tsinghua University Volunteer Public Welfare Scholarship.
- Competitions Third Prize, Beijing 36th Undergraduate Mathematics Competition; Third Prize, 2025 Chinese Mathematics Competition for College Students, Beijing Division; Third Prize, 2025–2026 Tsinghua Theoretical Mechanics Competition.
- Innovation iFLYTEK Xingchen Agent Design Excellent Creativity Second Prize; Tencent Hunyuan 3D Model Design Excellent Creativity First Prize; Tsinghua Xingjian College “Wuxing Cup” Most Feasible Award.
- Other 2024 undergraduate military-training advanced individual; 2021–2022 Hebi Senior High School outstanding student; 2022 Hebi “Three-Good Student”; 2023 provincial second prizes in mathematics, physics, and biology competitions, and third prize in chemistry competition.